Junwei Huang

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Research Interests: Power management ICs (PMICs); high current density; fast transient response; chiplet/ vertical power delivery

EDUCATION

2018.08–2024.07	University of Macau	Electrical and Computer Engineering	Ph.D. Advisors: Prof. Yan Lu and Prof. Chi-Seng Lam
2014.09–2018.06	University of Electronic Science and Technology of China	Microelectronics Science and Engineering	B.Eng.

ACADEMIC APPOINTMENTS

2025.09–present	University of California, Berkeley	Department of Electrical Engineering and Computer Sciences (EECS)	Visiting Postdoctoral Researcher Faculty Host: Prof. Robert Pilawa-Podgurski
2025.09–present	University of Macau	Institute of Microelectronics	Postdoctoral Fellow PI: Prof. Rui P. Martins and Prof. Sai-Weng Sin
2024.11–2025.05	Tsinghua University	Department of Electronic Engineering	Visiting Scholar Faculty Host: Prof. Yan Lu
2024.10–2025.09	University of Macau	Institute of Microelectronics	Research Assistant PI: Prof. Rui. P Martins

RESEARCH SUMMARY

University of Macau, State Key Laboratory of Analog and Mixed-Signal VLSI **2018.08–present**

- My research focuses on the design of power management chips with high current density and fast transient response. To date, I have published 25 papers in international conferences and journals, including 17 conference papers at ISSCC (1 highlight paper as first author), CICC (3 paper as first author), and ISCAS, and 8 journals in JSSC (2 as first author), TCAS-I (1 as first author), OJ-SSCS, and JOS
- I was invited to present at an IEEE PES/PELS event held at UC Berkeley, recognized for expertise in PMIC design.

University of California, Berkeley, EECS **2025.09–present**

- My research focuses on the co-design of board-level power electronics and on-interposer high-voltage integrated voltage regulators (HV IVR, >6V input) for next-generation GPU/CPU power delivery, leveraging hybrid switched-capacitor/inductor converter topologies and scalable distributed converter arrays.

AWARDS AND ACADEMIC SERVICE

1. Akrostar Technology Academic Prize for the academic year 2023/2024 (Top 3).
2. Reviewer for IEEE Transactions on Industrial Electronics, IEEE Transactions on Circuits and Systems II.

RESEARCH WORK AND PUBLICATIONS

Conference:

[C1] **J. Huang**, X. Mao, Z. Tong, Z. Yu, W. Yang, C. -S. Lam, R. P. Martins, Y. Lu, "A 20MHz-1MHz Dual-Loop Non-Uniform-Multi-Inductor Hybrid DC-DC Converter with Specified Inductor Current Allocation and Fast Transient Response," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2025. **Highlight Paper**

[C2] **J. Huang**, Z. Tong, X. Mao, C. -S. Lam, R. P. Martins and Y. Lu, "A Fast-Slow Two-Module DC-DC Solution with Transient and Efficiency Improvements for 2.5D/3D Integration," 2024 IEEE Custom Integrated Circuits Conference (CICC), Apr. 2024.

[C3] **J. Huang**, Z. Tong, Y. Lu, C. -S. Lam, R. P. Martins, "A 5V-to-0.5V Inductor-First Inductor-on-Ground Switched Capacitor Multi-Path Hybrid DC-DC Converter," 2023 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2023.

[C4] **J. Huang**, X. Mao, Z. Tong, Z. Yu, S. Han, C. -S. Lam, R. P. Martins, Y. Lu, "A Scalable Distributed 12-to-1V Fast-Slow Hybrid DC-DC Combo with Fast-Transient Bucklets and One-Pin Current Balancing," 2026 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2026. **Accepted**

[C5] Z. Tong, **J. Huang**, Y. Lu, R. P. Martins, "A 42W Reconfigurable Bidirectional Power Delivery Voltage-Regulating Cable," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2023. **Highlight Paper**

[C6] Z. Zhu, **J. Huang**, Z. Song, Z. Yu, S. Han, S-W. Sin, Y. Lu "An Inductor-at-Middle Hybrid Buck Converter with Shared Power-Signal Path for Distributed Vertical Power Delivery," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2026. **Accepted**

[C7] Z. Yu, Z. Tong, **J. Huang**, J. Yin, Y. Lu "A Compact Dual-Capacitor Relay SPT Supply Modulator with Overshoot-Free Adaptive On-Time Control for 5G FR2 CMOS PA," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2026. **Accepted**

[C8] Z. Tong, Z. Yu, **J. Huang**, X. Mao, Bernhard Wicht, Rui P. Martins, Yan Lu, "HOOP: A Scalable Hybrid DC-DC Converter Ring for High Performance Computing," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2025.

[C9] W. Yang, Z. Tong, **J. Huang**, R. P. Martins, Y. Lu, "A Bi-Directional Dual-Path Boost-48V-Buck Hybrid Converter for High-Voltage Power Transmission Cable in Light-Weight Humanoid Robots" in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2025.

[C10] Z. Yu, **J. Huang**, Z. Tong, Z. Zhu, Y. Lu, "An Output-Rail-Scalable Hybrid Buck Converter with Globally Adaptive Frequency Control Achieving 98.0% Peak Efficiency and 490 mW/mm3 Power Density," 2026 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2026. **Accepted**

[C11] S. Han, **J. Huang**, F. Song, Z. Tong, S-W Sin, H. Jiang, X. Wu, T. Ren, Y. Lu, "A 94.1% Peak Efficiency Coupled-Inductor Boost Converter with Small Inductor Current Ripple and Fast Transient Response," 2026 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2026. **Accepted**

[C12] Z. Song, Z. Tong, **J. Huang**, Z. Zhu, S-W Sin, Y. Lu, "A 6V-to-1V ReSC-Hybrid Two-Stage Merged Buck Converter with an Ultra-Small Inductor for VPD," 2026 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2026. **Accepted**

[C13] X. Mao, **J. Huang**, Z. Tong, R. P. Martins and Y. Lu, "A Quad-Output Hybrid Buck Converter with 8-Inductor Helping One Spot from All Quarters for Multi-Core XPUs," 2024 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2024.

[C14] Z. Tong, W. Yang, S. Han, **J. Huang**, X. Mao, Y. Lu, "Where is the Inductor: A Review and Comparison of the Hybrid DC-DC Buck Topologies," 2025 IEEE Custom Integrated Circuits Conf. (CICC), April 2025.

[C15] Z. Yu, **J. Huang**, Z. Tong, M. Huang and Y. Lu, "An Always Dual-Path Hybrid DC-DC Converter with Multiphase Interleaving Switched-Capacitor Cell Obtaining 45% Output Ripple Reduction," IEEE International Symposium on Circuits and Systems (ISCAS), May 2025.

[C16] F. Luo, **J. Huang**, M. Huang and Y. Lu, "A 12V-Input 1.8V-0.8V-Output Multiple-Output Hybrid Buck DC-DC Converter with a Shared Flying Capacitor," IEEE International Symposium on Circuits and Systems (ISCAS), May 2025.

[C17] Y. Hu, **J. Huang**, M. Huang and Y. Lu, "A 5V-to-0.8V Inductor-First 2L2C Multi-Path Hybrid DC-DC Converter," IEEE International Symposium on Circuits and Systems (ISCAS), May 2025.

Journal:

[J1] **J. Huang**, Z. Tong, X. Mao, C. -S. Lam, R. P. Martins and Y. Lu, "A Fast-Slow Two-Module High-Power-Density DC-DC Solution With Transient and Efficiency Improvements," in IEEE Journal of Solid-State Circuit (JSSC), 2025.

[J2] **J. Huang**, Z. Tong, C. -S. Lam, X. Mao, R. P. Martins, Y. Lu, "A Multi-Path Inductor-First Inductor-on-Ground Switched-Capacitor Hybrid DC-DC Converter," in IEEE Journal of Solid-State Circuit (JSSC), 2024.

[J3] **J. Huang**, C. -S. Lam, Y. Lu and R. P. Martins, "A Symmetrical Double Step-Down Converter with Extended Voltage Conversion Ratio," in IEEE Trans. Circuits Syst. I: Regular Papers (TCAS-I), 2022.

[J4] Z. Tong, **J. Huang**, X. Mao, R. P. Martins and Y. Lu, "A Double Pulse Overlapping Laser Diode Driver with Minimum 100-ps Pulse for LiDAR System", in IEEE Journal of the Solid-State Circuits (JSSC), 2024.

[J5] Z. Tong, **J. Huang**, X. Mao, R. P. Martins and Y. Lu, "A Bidirectional USB Power Delivery Voltage-Regulating Cable", in IEEE Journal of the Solid-State Circuits (JSSC), 2024.

[J6] W. Yang, Z. Tong, **J. Huang**, R. P. Martins and Y. Lu, "A Bi-Directional Dual-Path Boost-Buck Hybrid Converter for High-Voltage Power Transmission Delivery Cable in Humanoid Robots," in IEEE Journal of the Solid-State Circuits (JSSC), 2025.

[J7] Y. Lu, **J. Huang**, Z. Tong, T. Hu, W.-L. Zeng, M. Huang, X. Mao and G. Cai, "An Overview of Hybrid DC-DC Converters: From Seeds to Leaves," in IEEE Open Journal of the Solid-State Circuits Society (OJSSC), 2024.

[J8] Y. Lu, G. Cai, and **J. Huang**, Favorable basic cells for hybrid DC-DC converters[J]. Journal of Semiconductors (JOS). 2023.